



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to CKM FUELS STATION

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

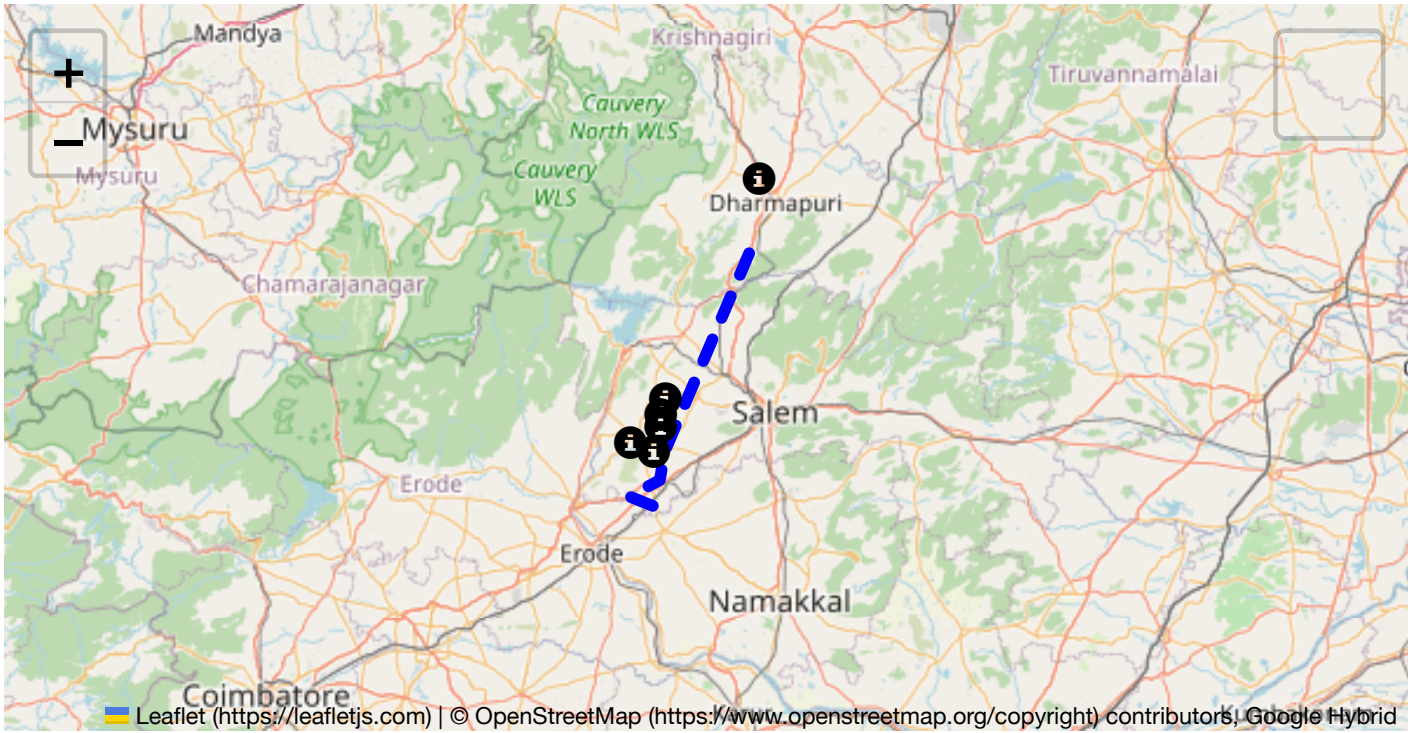
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 96.49 km
Estimated Duration: 2.1 hours
Adjusted Duration (Heavy Vehicle): 2.6 hours

Start: (11.4381, 77.8734)
End: (12.066107, 78.116553)



Welcome to the Journey Risk Management Study

Route Overview

The route from Sangagiri to Dharmapuri spans approximately 96.49 kilometers and typically takes around 2.11 hours for heavy vehicles. The journey includes key locations such as Bhavani Main Road, Ayveli on SH 86, Olakkachinnanur, and Erumaipatti. The roads predominantly traverse rural and semi-urban areas, with stretches of both highway and local roads.

Typical Weather Conditions

Tamil Nadu generally experiences a tropical climate. From June to September, the region is subject to the Southwest monsoon, leading to heavy rains, which can cause poor road conditions, especially in rural segments. The period from October to December sees the Northeast monsoon, with the potential for cyclones. Dry months are typically from January to May, with occasional heat waves.

Traffic Patterns

Traffic congestion can be expected near major town intersections and entry points, particularly during morning (8:00 AM - 10:00 AM) and evening (5:00 PM - 7:00 PM) peak hours. Notably congested areas include Bhavani Main Road and intersections around Ayveli.

Road Quality and Infrastructure

The National Highways and State Highways along this route are generally well-maintained. However, rural roads may be narrower and less well-maintained, with occasional potholes and uneven surfaces. Poorly lit segments at night can pose visibility challenges.

Alternative Routes for Emergencies

For emergencies, an alternative could be rerouting through the NH44 for more direct access to Dharmapuri, though it may involve extended travel distance. Local rural paths should be used cautiously and preferably with local guidance due to varying road conditions.

Local Regulations on Hazardous Material Transport

Transporting hazardous materials in Tamil Nadu requires adherence to national safety standards, including permits, proper labeling, and emergency procedure protocols. Night travel for such goods might be restricted in certain regions.

Historical Incidents

Historically, minor accidents involving heavy vehicles are reported occasionally owing to narrow rural roads and unforeseen weather conditions, with major incidents being rare.

Environmental Considerations

The route crosses proximity to agricultural zones. Care should be taken to prevent contamination, particularly as waterways and sensitive ecological zones like forest patches may be encountered.

Communication Coverage

While urban and many rural segments have good mobile network coverage, certain stretches, especially in remote rural areas, may have communication dead zones. Drivers should ensure alternative communication means or pre-communication at check-in points.

Estimated Emergency Response Times

In urban and semi-urban areas, emergency response can take 20-30 minutes, while in remote rural areas, it may extend to over an hour. Pre-identifying hospitals and emergency services along the route is advisable.

Risk Assessment Summary

Overall, the route is moderately safe, with increased risks during the monsoon season due to weather and potential road degradation. Traffic congestion and narrow rural roads are the primary concerns. Drivers should adhere to local regulations rigorously, maintain communication readiness, and be aware of environmental sensitivity to mitigate risks effectively. Regular updates on road conditions and weather forecasts are recommended for optimal safety.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43968, 77.87345	15 KM/Hr
1	Turn	High	11.44029, 77.87544	15 KM/Hr
2	Turn	Medium	11.44900, 77.87406	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
3	Turn	High	11.45353, 77.85697	15 KM/Hr
4	Turn	Medium	11.45853, 77.85139	30 KM/Hr
5	Turn	Medium	11.45906, 77.85132	30 KM/Hr
6	Turn	High	11.46307, 77.84941	15 KM/Hr
7	Turn	High	11.45542, 77.81725	15 KM/Hr
8	Turn	High	11.45636, 77.81678	15 KM/Hr
9	Turn	Medium	11.49383, 77.88501	30 KM/Hr
10	Blind Spot	Blind Spot	11.49400, 77.88538	10 KM/Hr
11	Turn	High	11.49469, 77.88488	15 KM/Hr
12	Turn	Medium	11.56083, 77.89921	30 KM/Hr
13	Turn	Medium	11.58309, 77.90652	30 KM/Hr
14	Turn	Medium	11.63843, 77.94821	30 KM/Hr
15	Turn	High	11.64192, 77.95195	15 KM/Hr
16	Turn	Medium	11.64747, 77.95129	30 KM/Hr
17	Turn	High	11.67868, 77.96836	15 KM/Hr
18	Turn	High	11.70223, 77.97860	15 KM/Hr
19	Turn	Medium	11.71380, 77.99532	30 KM/Hr
20	Turn	Medium	11.71731, 78.00006	30 KM/Hr
21	Turn	Medium	11.71947, 78.01172	30 KM/Hr
22	Turn	Medium	11.72760, 78.02285	30 KM/Hr
23	Turn	Medium	11.73785, 78.03496	30 KM/Hr
24	Turn	High	11.73653, 78.03886	15 KM/Hr
25	Turn	Medium	11.74252, 78.04705	30 KM/Hr
26	Turn	Medium	11.87998, 78.08170	30 KM/Hr
27	Turn	Medium	11.91266, 78.05852	30 KM/Hr
28	Turn	Medium	11.93746, 78.05331	30 KM/Hr
29	Turn	Medium	11.95803, 78.04964	30 KM/Hr
30	Turn	Medium	11.95854, 78.04990	30 KM/Hr
31	Turn	Medium	11.96015, 78.04945	30 KM/Hr
32	Turn	Medium	11.96370, 78.05142	30 KM/Hr
33	Turn	Medium	11.96433, 78.05113	30 KM/Hr
34	Turn	High	11.96542, 78.04938	15 KM/Hr
35	Turn	High	11.96650, 78.04927	15 KM/Hr
36	Turn	Medium	11.96804, 78.05080	30 KM/Hr
37	Turn	High	12.05835, 78.10789	15 KM/Hr
38	Turn	Medium	12.05840, 78.10812	30 KM/Hr
39	Blind Spot	Blind Spot	12.05721, 78.11107	10 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Sri Shanthi Hospital	11.569272, 77.901937	30 km/h	Medium
2	hospital	Sugumar Hospital	11.7396031, 78.0426955	30 km/h	Medium
3	hospital	Omalur Government Hospital	11.74161, 78.044296	30 km/h	Medium
4	clinic	R.R. Nursing Home	11.7394129, 78.0439861	30 km/h	Medium
5	hospital	Sri Hari Hospital	11.7396726, 78.0440701	30 km/h	Medium
6	hospital	Saravana Hospital	11.7409303, 78.0458308	30 km/h	Medium
7	clinic	Saratha Clinic	11.7422818, 78.0458175	30 km/h	Medium
8	hospital	Sri Sugam Hospital	11.7429821, 78.0461039	30 km/h	Medium
9	hospital	Arun Hospital	11.7429892, 78.0466881	30 km/h	Medium
11	clinic	Sri Meenakshi Nursing Home, Salem	11.7449162, 78.0474637	30 km/h	Medium
12	clinic	Divyam Child Care Clinic	11.7447135, 78.0473858	30 km/h	Medium
13	clinic	Momota Clinic	11.744982, 78.047397	30 km/h	Medium
14	clinic	Gomathi Clinic	11.7445293, 78.0470401	30 km/h	Medium
15	hospital	Vinayaka Child Hospital	11.745812, 78.0480225	30 km/h	Medium
16	hospital	Dr. G.R.Thilakam Hospital	11.746097, 78.04769	30 km/h	Medium
17	hospital	Amirthalingam Hospital	11.7456708, 78.0470548	30 km/h	Medium
18	hospital	Children Hospital	11.7458377, 78.0480205	30 km/h	Medium
19	hospital	Dr. Nagaraj Hospital	11.7464035, 78.0472401	30 km/h	Medium
20	hospital	Thulasi Hospital	11.7461516, 78.0469984	30 km/h	Medium
21	clinic	Shri Ramana Poly Clinic	11.7477122, 78.0479293	30 km/h	Medium
22	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
23	hospital	St. Joseph's Speciality Hospital	11.761611, 78.0522344	30 km/h	Medium
25	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
27	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
28	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
29	hospital	Thoppur,Government Hospital	11.9398824, 78.0551658	30 km/h	Medium
30	clinic	Seva Clinic	12.0609666, 78.1130428	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
10	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
24	school	John Britto Matriculation Higher Secondary School	11.7676907, 78.0513092	30 km/h	Medium
26	college	Dhirajlal Gandhi College of Technology	11.7968014, 78.0562095	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45353, 77.85697



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45853, 77.85139



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45906, 77.85132



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46307, 77.84941



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45542, 77.81725



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45636, 77.81678



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49383, 77.88501



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.49400, 77.88538



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49469, 77.88488



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.56083, 77.89921



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.58309, 77.90652



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.63843, 77.94821



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.64192, 77.95195



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64747, 77.95129



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.67868, 77.96836



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.70223, 77.97860



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71380, 77.99532



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71731, 78.00006



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71947, 78.01172



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.72760, 78.02285



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.73785, 78.03496



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.73653, 78.03886



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.74252, 78.04705



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.87998, 78.08170



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.91266, 78.05852



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93746, 78.05331



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95803, 78.04964



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95854, 78.04990



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96015, 78.04945



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96370, 78.05142



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96433, 78.05113



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96542, 78.04938



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96650, 78.04927



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96804, 78.05080



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.05835, 78.10789



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.05840, 78.10812



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 12.05721, 78.11107
